

Zachary Arnold

Mathematics & Data Science Student | Cybersecurity Analyst | Quantitative Research

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SUMMARY

Mathematically-driven freshman at UMSL with hands-on experience in cybersecurity, client-facing technical consulting, and quantitative research. Financially independent first-generation college student who has managed multi-department operations, delivered professional security assessments, and authored a 35-page quantitative trading case study. Seeking roles in data analysis, software development, or quantitative finance where analytical skills and real-world technical experience create immediate impact.

EDUCATION

B.S. Mathematics — Data Science Emphasis

Aug 2026 – Present

University of Missouri–St. Louis (UMSL) • St. Louis, MO

- Enrolled full-time (15 credit hours); coursework includes College Algebra, Trigonometry, and Composition.
- Pursuing CLEP Calculus credit (Dec 2026) to accelerate into upper-division quantitative coursework.

High School Diploma

Graduated May 2026

Mexico Senior High School • Mexico, MO

- Completed dual-enrollment Elementary Statistics at Moberly Area Community College (B); transferred to UMSL as MATH 1105.
- Completed Programming I & II and an independent study in Data Science (Excel, SQL, data analysis fundamentals).

EXPERIENCE

Cybersecurity Analyst & Sales

Apr 2026 – Present

Expert Tech Solutions (Remote) •

- Assessed and documented cybersecurity vulnerabilities for 3 small-business clients; delivered remediation steps for each.
- Produced 2 client-ready security assessment reports detailing vulnerabilities, risk levels, and step-by-step remediation plans.
- Led a full office hardware migration for 1 client site — decommissioned, relocated, and reinstalled all equipment with zero downtime on cutover day.

Entertainment Team Associate

Early 2026 – Present (EDC Leave)

Walmart Store #25 • Mexico, MO

- Managed 3 departments simultaneously (Electronics, Stationery, Paint Booth); workload required 2 replacement hires after departure.
- Averaged 40–45 hours per week while scheduled part-time; one of only 5 associates cross-trained on the paint booth.

Volunteer — Janitorial & Facilities Support

Summer 2025 – Dec 2025

YMCA & Local Senior Center • Mexico, MO

- Maintained 2 community facilities (YMCA and Senior Center) serving 50–150 members daily.
- Volunteered 2+ semesters supporting underserved youth and senior populations; consistently recognized for reliability.

PROJECTS

Quantitative Trading Case Study

2026

Independent Research •

- Backtested 50/200-day SMA crossover vs. buy-and-hold on 5 large-cap equities (AAPL, MSFT, JPM, XOM, JNJ) across 2010–2023.
- Implemented full data pipeline in Python: data ingestion, signal generation, Sharpe ratio (0.779 vs 0.849 B&H;), max drawdown, and t-test validation.
- Published on SSRN (DOI: 10.2139/ssrn.6684499) — 172 downloads, 379 views.

TECHNICAL SKILLS

Languages: Python, HTML/CSS, SQL

Tools: Git, VS Code, Excel, Google Sheets

Data & Quant: Pandas, NumPy, backtesting, SMA crossover strategy, statistical analysis

Cybersecurity: Vulnerability assessment, client security consulting, threat documentation

Other: Experience building web applications; formal and independent coursework in mathematics and statistics

LEADERSHIP & ACTIVITIES

- Drum Line — Mexico Senior High School: Informal section leader; peers consistently sought guidance on technique and execution; drove rehearsal cohesion.
- Putnam Mathematical Competition (Dec 2026): Awaiting registration opening; actively preparing in number theory, combinatorics, and algebraic manipulation.